

Internship Final Report

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Internship Duration: August 1st, 2026 – August 31st, 2026
Company: ShadowFox
Role: Data Analyst Intern
Mentor: Mr. Aakash

Objectives

The primary objective of this internship was to gain hands-on experience in data analysis using Microsoft Excel by working on a real-world retail sales dataset. Specific goals included learning how to integrate multiple related data files into a single analysis-ready table, clean and validate raw business data, calculate key performance metrics, and present findings through a clear, easy-to-read dashboard. Additional objectives included developing a habit of validating data before drawing conclusions, understanding how to distinguish between revenue-driving and profit-driving segments of a business, and building a practical understanding of how spreadsheet-based analysis supports real decision-making — such as inventory planning, promotional timing, and regional sales strategy. The internship was also intended to build foundational workplace skills, including working to a defined project structure, documenting findings clearly, and presenting results in a way that a non-technical stakeholder could understand at a glance.

Tasks and Responsibilities

As part of the Beginner-level track, I worked on the Retail Sales Analysis Dashboard project using the Madhav E-Commerce Sales Dataset. My key responsibilities included:

- Reviewing the Orders (500 records) and Details (1,500 records) files and integrating them into a single Sales_Data table using Order ID as the common key.
- Cleaning and standardizing the merged dataset — correcting data types, keeping Order ID as text, formatting Order Date, checking for missing values, and validating genuine duplicate records separately from expected repeated Order IDs.
- Organizing the Excel workbook into clearly separated sheets: Orders, Details, Sales_Data, Analysis, and Dashboard, to keep raw data, working data, and final output distinct.
- Calculating key summary metrics such as Total Sales, Total Profit, Total Quantity Sold, Total Orders, and Average Order Value, using unique Order ID counts rather than row counts.
- Performing monthly, category, sub-category, state-level, and payment-mode analysis to uncover sales patterns and identify strong and weak performing segments.
- Designing a final Dashboard sheet with KPI cards, four supporting charts, a Business Insights section, and a Recommended Actions section.
- Testing and validating the workbook for correct data types, chart source ranges, currency formatting, and overall dashboard readability before final submission.

Key Metrics Achieved

The final analysis produced the following headline results from the dataset:

Total Sales	Total Profit	Total Qty Sold	Total Orders	Avg Order Value
₹437,771	₹36,963	5,615	500	₹875.54

- Strongest month: January 2018 (~₹61,632); weakest month: July 2018 (~₹12,966)
- Top category by sales: Electronics (~₹166,267); top category by profit: Clothing (~₹13,325)
- Top sub-category: Printers (~₹59,252); leading state: Maharashtra (~₹102,498); most-used payment mode: COD (~2,456 units)

Learning Outcomes

This internship strengthened my practical Excel skills and my ability to think analytically about business data. I learned how to combine related tables correctly, including recognizing that repeated Order IDs were valid multi-line orders rather than duplicate errors — an important distinction that shapes how a dataset should be interpreted before any analysis begins.

I also gained experience validating data types and structure before drawing conclusions, calculating business metrics accurately (such as computing Total Orders from unique Order IDs rather than row counts), and using charts to identify meaningful patterns — for example, distinguishing between the highest-selling category and the most profitable one, which are not always the same. Finally, I learned how

to translate raw analysis into a simple, decision-ready dashboard that a business stakeholder could read at a glance, and how to document a project clearly enough for someone else to follow the reasoning behind each finding.

Challenges and Solutions

One early challenge was noticing that Order ID values repeated across the Details file, which initially looked like duplicate data. By reviewing the dataset structure carefully, I understood that this was expected, since a single order could include multiple product lines, and adjusted my analysis approach accordingly rather than incorrectly removing valid rows.

Another challenge was ensuring that summary metrics like Total Orders were calculated correctly; a simple row count would have overstated the true number of orders, so I used unique Order ID counts instead. I also had to be careful with data type formatting, particularly keeping Order ID as text and Order Date in a proper date format, to ensure accurate sorting and grouping during the monthly analysis. Finally, keeping the dashboard visually simple while still covering all required KPIs and charts required a few rounds of layout adjustment to avoid clutter.

Conclusion

Overall, this internship gave me a solid foundation in spreadsheet-based data analysis. Starting from raw, unprocessed order and transaction data, I was able to build a clean analysis-ready dataset and a functional Excel dashboard that clearly communicates sales performance, profitability, and regional and payment-mode trends. The project reinforced the importance of careful data validation before analysis and showed me how structured Excel work can directly support real business decisions such as inventory planning and promotional strategy. It has given me a strong practical base to build on as I take on more advanced data analysis work in the future.

Acknowledgments

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